Question				Marking details	Marks Available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
1	(a)			33 000 – 9 000 (1 for either figure given) = 24 000 [kilotonnes] (1) Award 1 mark for an answer of 24 [kilotonnes]		2		2	2	
	(b)			2008 **and** 2009		1		1	1	
	(c)			**Any 2 × (1) from:** • <u>Decreased</u> climate change / extreme weather • <u>Reduction</u> in global warming / greenhouse gases • <u>Reduction</u> in rising sea levels / melting icecaps • <u>Less danger</u> to habitats / reduction in extinction of some species N.B. Don't accept stop in any of the above Reference to pollution treat as neutral	2			2		
	(d)	(i)		Nuclear power stations <u>emit no CO_2</u> (1) so the line should be on 0 (1)			2	2		
		(ii)		Useful energy transferred = 185 (units) (1 – appearing anywhere) $= \frac{185}{500}$ [x 100] = 37[%] (1) so disagree **Alternative:** $\frac{315}{500}$ [x 100] = 63% Efficiency = 100 – 63 (1) = 37[%] (1) so disagree **N.B.** Award 1 mark for an answer of 0.37			2	2	2	

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				Alternative: 63% of 500 = 315 (1) This is wasted energy (1) so disagree **Award 2 marks for all of the following:** $\frac{315}{500}$ [x 100] = 63%, but this is the inefficiency (2) so disagree **Conclusion must be present to award 2 marks**						
				Question 1 total	**2**	**3**	**4**	**9**	**5**	**0**